REN SHAPE® 472

Machinable board for prototype tooling & assembly

REN Shape 472 is a polyurethane-based board designed for CNC machining of production grade checking and assembly fixtures. It may also be used in prototype thermoforming and vacuum forming tooling applications, providing excellent definition and dimensional stability at elevated temperatures. REN Shape 472 is a moderate density, neutral-colored board with excellent toughness. REN Shape 472 features the following properties:

- Ultra-fine surface finish
- High flexural strength
- Outstanding resistance to shrinkage and warping
- Tough and durable

Typical Properties

Color	Light grey
Density (ρ)	55 lb/ft^3 [881 kg/m^3]
Hardness	81 Shore D
Flexural Strength	9,540 psi [65.7 Mpa]
Flexural Modulus	416,000 psi [2868.2 Mpa]
Compressive Strength	10,900 psi [75.1 Mpa]
Compressive Modulus	460,000 psi [3171.5 Mpa]
Tensile Strength	7,000 psi [48.2Mpa]
Glass Transition Temp	248^{0}F [120^{0}C]
Coefficient of Thermal Expansion (CTE)	25.8 x 10^{-6} in/in^{0}F [46.4 x 10^{-6} m/m/^{0}C]

Available Sizes

1” x 20” x 60” (25.4mm x 508mm x 1524mm)
2” x 20” x 60” (51mm x 508mm x 1524mm)
3” x 20” x 60” (76.4mm x 508mm x 1524mm)
4” x 20” x 60” (101.6mm x 508mm x 1524mm)

Accessories

REN-Weld 103 for matched adhesives

Machining

Cutters: *Roughing* - 1”(25.4mm) Hog Ball End Mill, 4-flute, HS Steel 8% Cobalt
Finishing - 5/8”(15.875mm) Ball End Mill, 2-flute carbide
Depth: *Roughing* - varied from ¼”(6.35mm) to 2.5”(63.5mm) with 40% stepover
Finishing - 1/8”(3.175mm) deep leaving 0.002”(0.05mm) scallop height
Roughing speed: 1600RPM · Roughing Feed 40IPM
Finishing speed: 10,000RPM · Finishing feed 100IPM

Safety

Do not use/handle this product until the MSDS has been read and understood.

Applications

REN 472 applications may include prototype thermoform and vacuum form molds, foundry patterns and dimensionally stable tooling aids and fixtures.

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